Enhancing Local and Global Structure in Compressed AI Agent Memory

Henry Han

Abstract—AI agents store what they read as embedding vectors and compress them at scale. A compression is accepted when each item's nearest neighbors survive. Agents also consolidate, delete, and compare what they store, and these operations depend on which items lie far apart. We read both ends of a compressed memory from one chance-corrected locality profile: a local fidelity L at ten neighbors and a global fidelity G at 30 to 50 percent of the memory, both label-free. We prove that one minus the profile ν_k at every scale k is at most $N/\sqrt{k(N-k)}$, with $N = n - 1$ for n items, times the root rate of rank violations between near and far items. This factor lies between 2 and 2.18 across the band of G, so keeping near–far pairs in order keeps the far end. Locality-aware quantization (LAQ) turns the bound into a hinge objective. It learns the codebooks of an OPQ index and keeps its bytes, rotation, and search. On two agent memories and six encoders, retrieval follows L (partial rank correlation 0.83 to 0.99 given G). With the three retrieval-tuned encoders, consolidation and deletion follow G (0.55 to 0.93 given L). On held-out companies and conversations, LAQ raises G in all 40 runs with three retrieval-tuned encoders and Amazon Titan V2. It lifts the lowest theme agreement from 0.80 to 0.89 and has the highest G of seven quantizers at the same size in each of 12 runs.

Index Terms—Vector quantization, embedding compression, agent memory, neighborhood preservation, rank correlation, product quantization, label-free evaluation.

1 Introduction

Language-model agents remember by embedding what they read, say, and do. When they need an item, they recall the nearest embeddings [1]. Managed agent memories extract records from conversations, retrieve them by semantic search, and consolidate them over time [2]. At millions of vectors, every byte per vector costs memory, latency, and money. The vectors are therefore compressed: by truncating Matryoshka embeddings [3], by projection, by scalar or binary codes, or by product quantization [4]. Cloud services offer each of these as a switch [5, 6].

How should an agent memory be compressed so that it keeps both what lies near and what lies far? Today a compression is accepted by its nearest neighbors. Vendors report that binary embeddings keep 97 to 98.5 percent of retrieval accuracy [5], and they evaluate quantized indexes by recall@100 [6]. Studies of embedding compression report nDCG@10 [7] or benchmark task scores [8]. An agent also acts on what lies far apart. It consolidates a year of filings into themes [9, 10], deletes every item of a client or a session on request, and maps how companies and sectors relate. These operations compare each item with the whole memory, and they decide what a sector brief says and what a deletion removes. Two things make the far end hard to keep. A private agent memory has no labels to test it with, and quantizers are trained for reconstruction, nearest neighbors, or query relevance. Our idea is to measure the far end with the statistic that measures the near end, to prove which pairwise order controls it, and to train the codebooks on that order.

Figure 1 shows the two ends in a financial memory. Passages of one company form islands at the rim. In the outer fifth of the map, 94 percent of each passage's 10 nearest map neighbors come from its own company. Passages of many companies meet in shared themes at the center. At 96 bytes per passage, rescored binary codes keep chance-corrected recall@10 at 0.97 and move 10 percent of the passages to another theme. PCA with 8-bit codes moves 2 percent.

Measurement. We read a chance-corrected locality profile [12, 13] at its two ends. The local fidelity L is chance-corrected recall@10. The global fidelity G averages, over 30 to 50 percent of the memory, the agreement on which items lie in each item's nearer part. At its upper edge the profile is the agreement on near and far halves, and under a Gaussian copula it equals Kendall's τ of the distances. Both fidelities need only the vectors before and after compression. G is estimated from a subsample of 2,000 items, however large the memory.

Learning. We prove that rank violations among cross pairs control the profile at every scale. A cross pair is a near and a far item of one anchor. Let ν_k be the profile at scale k and $N = n - 1$ for a memory of n items. Then $1 - \nu_k$ is at most $N/\sqrt{k(N-k)}$ times the root of the violation rate. The factor is between 2 and 2.18 across the band of G. A hinge on distance gaps bounds the violation rate. Locality-aware quantization (LAQ) lowers this hinge at middle and far scales by learning the codebooks of an optimized product quantizer (OPQ) [14]. It needs no labels or queries. The PCA basis, the rotation, the bytes per item, and the search stay those of the OPQ index.

Contributions. In order of importance, each linked to the next:

1) *A quantizer that keeps the far end.* LAQ learns product-quantization codebooks that keep middle- and far-scale rank order. It is label-free and replaces an OPQ index at the same bytes and search.

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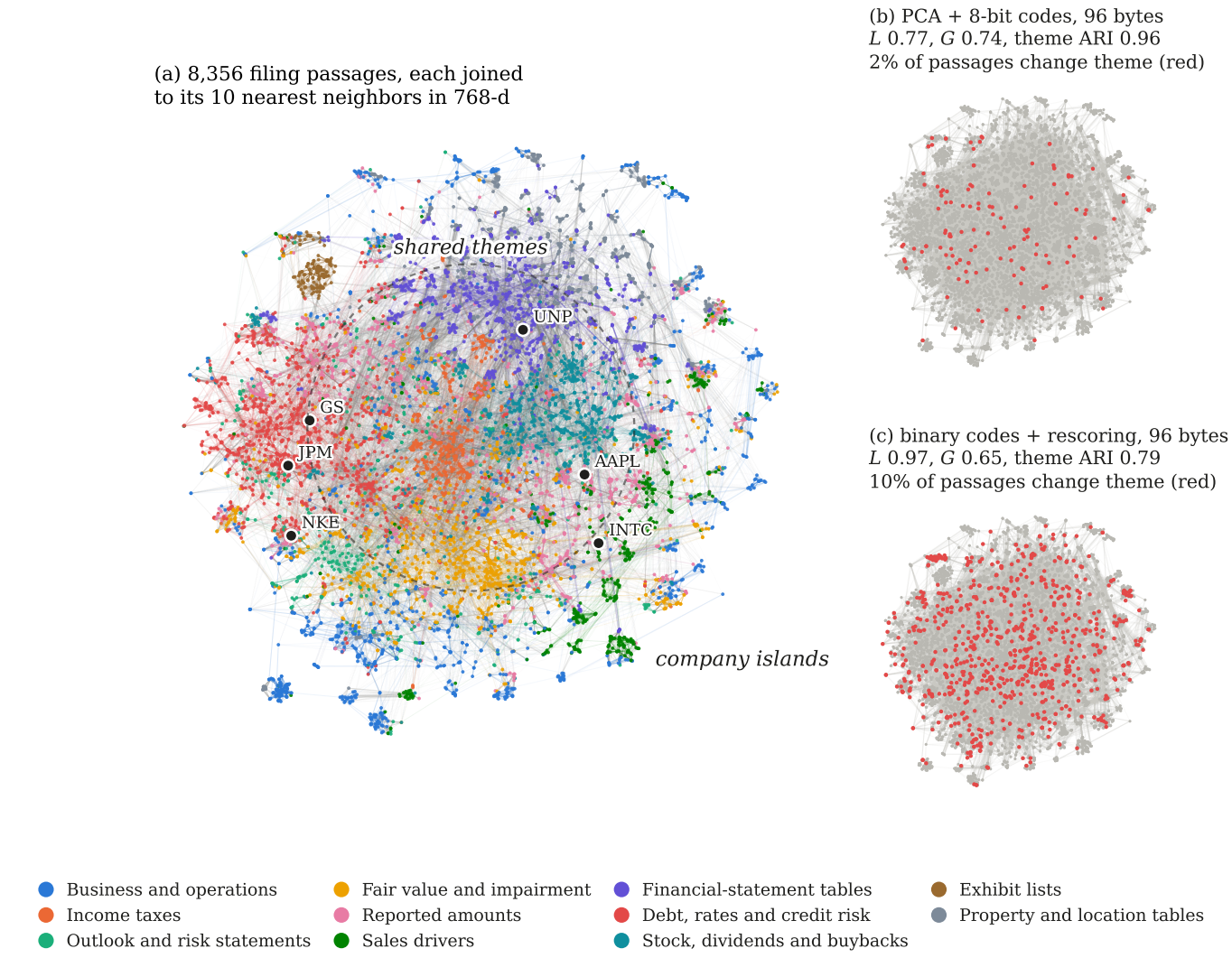


Fig. 1. Two compressions of equal size keep the near end alike and the far end differently. FinQA, BGE-base, 8,356 passages of 106 S&P 500 companies, t-SNE layout [11]. (a) Each passage is joined to its 10 nearest neighbors and colored by one of 11 themes of the uncompressed memory. (b, c) At 96 bytes per passage, rescored binary codes keep more neighbors than PCA with 8-bit codes (L 0.97 against 0.77) and move more passages to another theme (red; 10 against 2 percent).

2) *The bound behind it.* A rank-violation bound ties the profile at every scale to the order of cross pairs, and a hinge surrogate makes the bound trainable. Both come with proofs, together with a self-contained account of the two-end profile.
3) *Held-out evidence.* On held-out companies and conversations of three memories, with three retrieval-tuned encoders and Amazon Titan V2, LAQ raises G in all 40 runs and L in 34. It lifts the lowest theme agreement on Multi-Doc-2025 from 0.80 to 0.89 and has the highest G of seven quantizers at the same size.
4) *Why the far end matters.* Each end governs its own operations. On two agent memories and six encoders, retrieval follows L (partial rank correlation 0.83 to 0.99 given G). With the retrieval-tuned encoders, consolidation and deletion follow G (0.55 to 0.93 given L). Rescoring repairs the near end and leaves the far end of the stored codes.

A compressed memory can now be tested at both ends and trained to keep both. The same profile and objective extend to any vector index whose operations read beyond the nearest neighbors, such as clustering, deduplication, and topic maps.

What is new. The closest quantizers learn codebooks with pairwise or triplet losses. PQN uses triplets from class labels [15]. Spreading vectors pair a triplet loss on nearest neighbors with an entropy term [16]. DPQ trains product codes end to end through straight-through assignments [17], and JPQ and Distill-VQ train with queries and relevance labels [18, 19]. LAQ differs in target, input, and output. Its target is rank order at 30 to 50 percent of the memory, chosen by a bound. Its input is the memory alone. Its output changes only the codebook values of an OPQ index. On the measurement side, the profile is the R_{NX} curve of Lee and Verleysen [20], and our earlier work defined L and G on it for two-dimensional maps [13]. Here the profile measures compressed high-dimensional memories, and the rank-violation bound is new.

Section 2 reviews related work. Section 3 gives the

measurement theory and the bound, and Section 4 the quantizer. Section 5 describes the memories, encoders, and operations. Section 6 shows what each end measures, and Section 7 tests LAQ on held-out memories.

2 Related work

Vector quantization. Product quantization (PQ) splits a vector into subvectors and codes each separately [4]. OPQ learns a rotation that balances the subvectors [14]. Residual quantization (RQ) [21] and local search quantization (LSQ++) [22] code a vector as a sum of codewords. ScaNN weights the quantization error by its effect on inner products with queries [23]. Iterative quantization (ITQ) rotates the data before taking signs [24], and RaBitQ gives randomized binary codes with an error bound on distance estimates [25]. Learned quantizers train codebooks end to end. PQN uses an asymmetric triplet loss on labeled images [15]. UNQ trains a nonlinear encoder for nearest-neighbor search [26]. Spreading vectors combine a triplet loss on nearest neighbors with an entropy term [16]. DPQ makes product quantization differentiable with soft assignments and straight-through codes [17]. JPQ, RepCONC, and Distill-VQ train the quantizer of a dense retriever with queries and relevance labels or with a teacher's rankings [18, 19, 27].

Evaluation of compressed vectors. Quantized indexes are evaluated by nearest-neighbor recall [6, 28], retrieval quality [5, 7], or task scores [8]. Kisako et al. [8] conjecture that clustering and retrieval depend on neighborhood geometry. Kuffo and Boncz [29] find that one-bit codes support the clustering an index uses to partition its vectors. Turaga et al. [30] showed that a moment-preserving quantizer per cluster keeps a given k-means partition. Our measure asks whether the compressed memory keeps the partition and the global arrangement of the uncompressed one, with no partition given in advance.

Local and global structure. Rank-based quality criteria compare neighborhoods across scales [20, 31]. Lueks et al. [32] split the co-ranking curve into local and global parts, and DREAMS and CMET balance or score both ends of visualizations [33, 34]. The global structure of embeddings is also scored by the correlation of distances and by triplet accuracy [35, 36]. We introduced the degree of locality preservation [12]. We then defined L and G on its chance-corrected profile for two-dimensional maps, with the subsample estimator of G and the link of its midpoint to Kendall's τ [13]. This paper carries them to compressed high-dimensional memories, adds the rank-violation bound, and turns the bound into a quantizer.

Agent memory. Benchmarks such as LoCoMo [37] and LongMemEval [38] test whether agents recall facts from long interactions. Managed memories consolidate new records against the most similar stored ones [2]. Corpus-level organization clusters embeddings into themes and summaries [9, 10, 39]. These systems compress text by summarizing and consolidating it. We study the compression of the vectors beneath them and the operations that read the whole memory.

3 Two-end fidelity of a compressed memory

One chance-corrected profile measures both ends of a compressed memory, and the order of near and far items controls it at every scale. This section defines the profile and proves what its two ends measure. It then defines the local and global fidelities L and G and proves the bound that LAQ optimizes in Section 4.

3.1 Setting

A memory holds n items with vectors $x_1, \dots, x_n \in \mathbb{R}^d$. A compression maps each x_i to a stored code. We write z_i for the vector the memory computes distances with. It is the decoded vector for scalar, PCA, and product codes, and the ± 1 vector for binary codes, compared by Hamming distance. Write $N = n - 1$. For item i, let $r_i^x(j) \in \{1, \dots, N\}$ be the rank of j among the other items by increasing distance to x_i, with ties broken by index. Let $r_i^z(j)$ be the rank among the compressed vectors. For $1 \le k \le N-1$ the k-neighborhoods are $\mathcal{N}_k^x(i) = \{j : r_i^x(j) \le k\}$ and $\mathcal{N}_k^z(i) = \{j : r_i^z(j) \le k\}$.

3.2 The locality profile

Definition 1 (Locality profile)**.** *The degree of locality preservation at scale k is*

$$\mathrm{DLP}_k = \frac{1}{nk} \sum_{i=1}^{n} \left| \mathcal{N}_k^x(i) \cap \mathcal{N}_k^z(i) \right|, \tag{1}$$

and its chance-corrected value is

$$\nu_k = \frac{N \,\mathrm{DLP}_k - k}{N - k}. \tag{2}$$

The locality profile is the curve $k \mapsto \nu_k$, read over the relative scale k/n.

DLP_k was introduced in our work on the explainability of manifold learning [12]. It equals the neighborhood-preservation criterion $Q_{NX}(k)$ of Lee and Verleysen [31], and ν_k equals their rescaled criterion $R_{NX}(k)$ [20]. At one k and with the uncompressed memory as reference, DLP_k is the recall@k by which approximate nearest-neighbor indexes are evaluated [28]. The profile reads this quantity at every scale, and the results below give its two ends a defined meaning. Propositions 1–3 and Definition 2 are from our earlier work on nonlinear embeddings [13]; we give the proofs so that the paper is self-contained.

In words, a compression that keeps neighbors at random scores 0, and one statistic serves codes whose distances live on different scales.

Proposition 1 (Chance level and invariance)**.** *(i) If each $\mathcal{N}_k^z(i)$ is a uniformly random k-subset of the other N items, independent of x, then $\mathbb{E}\,\mathrm{DLP}_k = k/N$ and $\mathbb{E}\,\nu_k = 0$. A compression that keeps every ranking has $\nu_k = 1$. (ii) ν_k is invariant under any rotation, reflection, translation, or uniform scaling of the z_i, and under any strictly increasing transformation of either distance.*

Proof. (i) Each of the k members of $\mathcal{N}_k^x(i)$ lies in a uniform k-subset of N items with probability k/N, so $\mathbb{E}|\mathcal{N}_k^x(i) \cap \mathcal{N}_k^z(i)| = k^2/N$. Substitution in (2) gives 0. (ii) Each of these operations keeps every ranking r_i^z. □

Invariance (ii) lets one statistic compare Euclidean distance between decoded vectors, Hamming distance between bits, and distances after a learned rotation. All 486 compressions of Section 6 lie above the chance level. The lowest

G among them, 0.04, belongs to UMAP to 16 dimensions (MobileCLIP2, FinQA).

The next result says what the profile reads at its midpoint: for every item, which half of the memory is near.

Proposition 2 (Near–far duality)**.** *Let $\mathcal{F}_k^x(i)$ and $\mathcal{F}_k^z(i)$ be the complements of $\mathcal{N}_k^x(i)$ and $\mathcal{N}_k^z(i)$ among the other N items. Then $|\mathcal{F}_k^x(i) \cap \mathcal{F}_k^z(i)| = N - 2k + |\mathcal{N}_k^x(i) \cap \mathcal{N}_k^z(i)|$. For $k^\star = N/2$ with N even,*

$$\nu_{k^\star} = 2A - 1,$$
$$A = \frac{1}{nN} \sum_{i=1}^{n} \Big| \{ j \neq i : \mathbb{1}[r_i^x(j) \leq k^\star] = \mathbb{1}[r_i^z(j) \leq k^\star] \} \Big|, \tag{3}$$

where A is the probability that the original and the compressed memory agree on whether a random other item lies in the nearer or the farther half.

Proof. Inclusion–exclusion gives the identity. At $2k^\star = N$ it reads $|\mathcal{F}^x \cap \mathcal{F}^z| = |\mathcal{N}^x \cap \mathcal{N}^z|$, so item i has $2|\mathcal{N}_{k^\star}^x(i) \cap \mathcal{N}_{k^\star}^z(i)|$ agreements out of N and $A = \mathrm{DLP}_{k^\star}$. Equation (2) with $N - k^\star = k^\star$ gives $\nu_{k^\star} = 2\mathrm{DLP}_{k^\star} - 1$. □

At small k the profile asks whether an item keeps its nearest neighbors. At $k^\star$ it asks, for every item, which half of the memory is near and which is far. The large distances decide that question, and so do the positions of themes, sessions, and companies relative to one another. In other words, the median split is a rank correlation of all distances to an item, as the next result makes exact under a Gaussian copula.

Proposition 3 (Median split and rank correlation)**.** *Fix item i and let u be a uniformly random other item. Suppose that, after strictly increasing transformations, the pair of distances $\big(\|x_i - x_u\|, \|z_i - z_u\|\big)$ is bivariate normal with correlation r_i. Then for large n the contribution of item i to $\nu_{k^\star}$ has expectation*

$$\mathbb{E}\,(2A_i - 1) = \frac{2}{\pi} \arcsin r_i, \tag{4}$$

which is Kendall's τ of the pair and a strictly increasing function of its Spearman correlation $\rho_i^S = \frac{6}{\pi} \arcsin \frac{r_i}{2}$.

Proof. The transformations map both medians to 0. For a standard bivariate normal pair with correlation r, Sheppard's formula [40] gives $\Pr(\text{both coordinates on the same side of } 0) = \frac{1}{2} + \frac{1}{\pi} \arcsin r$, so $\mathbb{E}(2A_i - 1) = \frac{2}{\pi} \arcsin r_i$. Kendall's τ and Spearman's ρ of the bivariate normal take the forms above [41], and both increase in r. □

The compressed memories follow this model closely. For each of the 486 compressions, we convert the Spearman correlation of all distances into $\frac{2}{\pi} \arcsin r$ through Proposition 3. The result has a rank correlation of 0.98 with the measured median split, and the two differ by 0.03 on average. The check pools all items of a memory, so it tests the model in aggregate.

3.3 Local and global fidelity

Definition 2 (Local and global fidelity)**.** *The local fidelity and the global fidelity of a compressed memory are*

$$L = \nu_{10}, \quad G = \frac{1}{|\mathcal{K}|} \sum_{k \in \mathcal{K}} \nu_k, \quad \mathcal{K} = \{k : 0.3 \leq k/n \leq 0.5\}. \tag{5}$$

L is chance-corrected recall@10. G averages the profile over the band whose upper edge is the median split of Proposition 2. Averaging over a band reduces the variance of a single $\nu_{k^\star}$, and it keeps the meaning of the median split. Within each of the 18 runs of Section 6, G and the median split have a rank correlation of 0.99 to 1.00 across the 27 compressions. Both scores are the same statistic at two scales and both are chance-corrected, so they are commensurable. A compression lies on the (L, G) plane, and its distance below the diagonal $G = L$ measures how much more of the far end it loses than of the near end.

Scale is defined relative to n, so G can be estimated from a subsample. If n_0 items are drawn uniformly, the $\lfloor \alpha n_0 \rfloor$ nearest within the subsample approximate the α-quantile neighborhood of the whole memory. We compute G on five uniform subsamples of 2,000 items and report the mean. On two-dimensional maps of up to 5,000 points, subsamples of 250 to 2,000 points estimate G within 0.003 of its full value [13]. L refers to the fixed scale $k = 10$ and is computed on the whole memory. All ν_k follow from the co-ranking matrix [31] by a two-dimensional cumulative sum, in $O(n_0^2 \log n_0)$ time.

Pipeline-aware reading. Each fidelity is read from the vectors its operations use. A retrieval pipeline can rescore: it fetches the 40 nearest codes and reranks them with full-precision vectors read from disk. Rescoring changes the lists a query returns and leaves the stored codes unchanged. We therefore read L and retrieval from the rescored lists, which the query sees. We read G and every other operation from the stored codes, which those operations run on.

Figure 2 shows the profile on the two agent memories. Scalar quantization to 8 bits keeps it near 1 at every scale. Four compressions to 96 bytes start at similar heights near $k = 10$ and separate as k grows. Rotated binary codes fall fastest toward the G band.

3.4 Rank violations bound the profile

The order of near and far items controls the profile at every scale. The profile counts neighbors kept, and a learning objective needs a quantity that changes smoothly with the codes. Pairwise rank order is such a quantity. For item i and scale k call (p, q) a *cross pair* if $p \in \mathcal{N}_k^x(i)$ and $q \notin \mathcal{N}_k^x(i)$: p is near and q is far in the original memory. Item i has $k(N - k)$ cross pairs. The pair is *violated* if the compression reverses it, $r_i^z(q) < r_i^z(p)$. Let $p_k(i)$ be the fraction of violated cross pairs of i and $\bar{p}_k = \frac{1}{n} \sum_i p_k(i)$.

The theorem below states that a memory cannot lose many neighbors at scale k unless it reverses many near–far pairs.

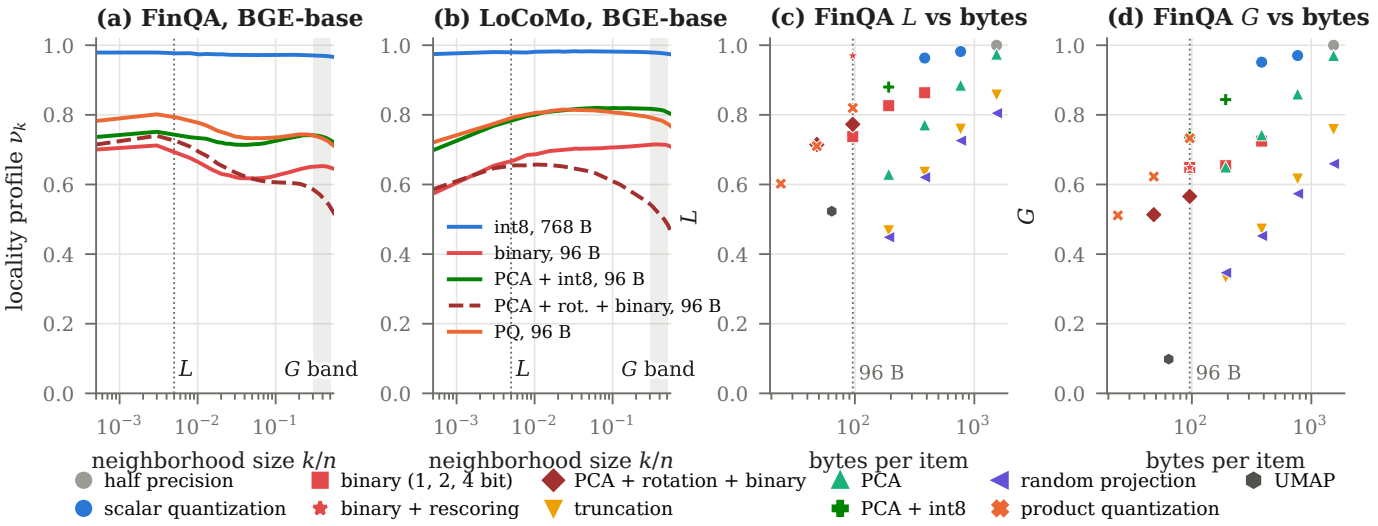


Fig. 2. Compressions of equal size start close at the near end and separate toward the far end. FinQA and LoCoMo, BGE-base. (a, b) Locality profiles of five compressions, one subsample of 2,000 items; dotted line: $k = 10$, where L is read; gray band: where G is averaged. (c, d) L and G of the 27 compressions of FinQA against bytes per item; at 96 bytes, G spreads from 0.57 to 0.74.

Theorem 1 (Rank-violation bound). *For every compression and every* $1 \le k \le N-1$,

$$1-\nu_k \;\le\; \frac{N}{\sqrt{k(N-k)}}\cdot\frac{1}{n}\sum_{i=1}^{n}\sqrt{p_k(i)} \;\le\; \frac{N}{\sqrt{k(N-k)}}\sqrt{\bar{p}_k}. \tag{6}$$

Proof. Fix i and let $m_i = k - |\mathcal{N}_k^x(i) \cap \mathcal{N}_k^z(i)|$. Exactly m_i items are *lost*, in $\mathcal{N}_k^x(i)$ and outside $\mathcal{N}_k^z(i)$, and exactly m_i are *gained*, in $\mathcal{N}_k^z(i)$ and outside $\mathcal{N}_k^x(i)$, because both sets have size k. Take a lost p and a gained q. Then p is near and q is far in the original memory, so (p, q) is a cross pair, and $r_i^z(q) \le k < r_i^z(p)$, so it is violated. These m_i^2 pairs are distinct, hence $m_i^2 \le p_k(i)\,k(N-k)$ and $m_i \le \sqrt{p_k(i)\,k(N-k)}$. By (1), $1-\mathrm{DLP}_k = \frac{1}{nk}\sum_i m_i$, and by (2), $1-\nu_k = \frac{N}{N-k}(1-\mathrm{DLP}_k)$. Combining,

$$1-\nu_k = \frac{N}{nk(N-k)}\sum_{i=1}^{n} m_i \le \frac{N}{\sqrt{k(N-k)}}\cdot\frac{1}{n}\sum_{i=1}^{n}\sqrt{p_k(i)}.$$

The second inequality is Jensen's inequality for the concave square root. □

The factor $N/\sqrt{k(N-k)}$ equals $1/\sqrt{(k/N)(1-k/N)}$. Across the G band, $k/N \in [0.3, 0.5]$, it lies between 2 and 2.18. A small violation rate at these scales therefore forces ν_k close to 1, and the bound transfers directly to G. At $k = 10$ the factor is about $\sqrt{N/10}$, which is 14 for a memory of 2,000 items. Theorem 1 thus singles out the middle and far scales as the scales where ordering cross pairs controls the profile, and LAQ trains its codes at those scales. On the held-out memories of Section 7, the bound holds at all 160 combinations of memory, quantizer, and scale, with $1-\nu_k$ at 0.49 to 0.60 of the bound.

Violations are counted by a step function, and a hinge on distance gaps bounds them. Let $\Delta_X = \|x_i - x_q\|^2 - \|x_i - x_p\|^2 \ge 0$ be the reference gap of a cross pair, $\Delta_Z = \|z_i - z_q\|^2 - \|z_i - z_p\|^2$ its compressed gap, and $h = [\eta\Delta_X - \Delta_Z]_+$ with margin $\eta > 0$. In plain terms, the corollary says that pushing compressed gaps above a fraction of the original gaps also pushes the violation rate down.

Corollary 1 (Hinge surrogate). *Let* (p, q) *be a uniformly random cross pair of a uniformly random item at scale* k*, and* $\bar{H}_k = \mathbb{E}\,h$*. For every* $\delta > 0$,

$$\bar{p}_k \;\le\; \Pr(\Delta_X < \delta) + \frac{\bar{H}_k}{\eta\,\delta}. \tag{7}$$

Proof. A violated pair has $\Delta_Z < 0$, so $h \ge \eta\Delta_X$. If also $\Delta_X \ge \delta$, then $h \ge \eta\delta$. Hence $\mathbb{1}[\text{violated}] \le \mathbb{1}[\Delta_X < \delta] + h/(\eta\delta)$, and taking expectations gives (7). Rankings by squared distance equal rankings by distance, so $\Delta_Z < 0$ for every violated pair. □

The first term depends on the original memory alone: it is the share of cross pairs whose two distances to i are within δ of each other. The second term is the expected hinge, and an unbiased estimate of it is the hinge averaged over cross pairs drawn uniformly. Theorem 1 and Corollary 1 together turn the profile at a chosen scale into an objective that gradient descent can lower. LAQ lowers this hinge, and on the held-out memories the violation rate at $k \approx 0.4n$ falls by 10 to 20 percent.

4 Locality-aware quantization

Locality-aware quantization (LAQ) trains the codebooks of an OPQ index to keep the order of cross pairs at the middle and far scales, the target that Theorem 1 names. Product quantizers usually learn codebooks to reconstruct vectors or to keep nearest neighbors. LAQ adds the far-end target, needs no labels, and keeps the rest of the OPQ index. Figure 3 shows the procedure.

Starting point. LAQ fits PCA on the training memory in the full basis. It then fits OPQ with $M = d/8$ subspaces of eight dimensions and $K = 256$ codewords each, so every item is stored in $d/8$ bytes. Let μ be the mean of the training vectors and R the orthogonal map that combines the PCA basis with

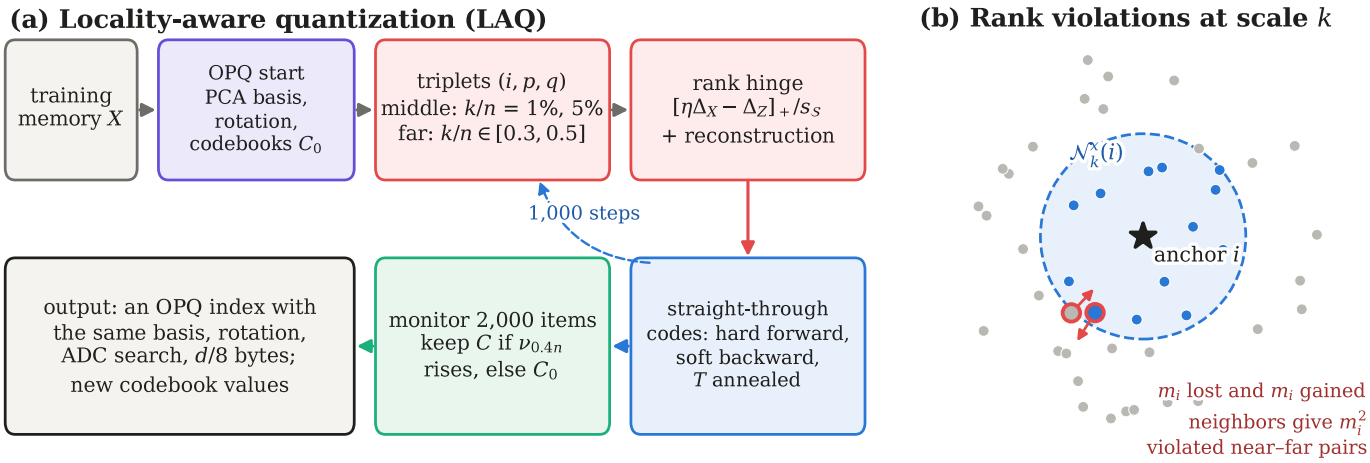


Fig. 3. LAQ changes only the codebook values of an OPQ index. (a) It samples cross pairs at middle and far scales, lowers a hinge on their order through straight-through codes, and keeps a checkpoint when the far end of a monitor set improves. (b) Why order controls the profile (Theorem 1): when anchor i loses m_i of its k nearest neighbors (blue) and gains m_i others (gray), every lost–gained pair is a violated cross pair.

the OPQ rotation. Let $y_i = R(x_i - \mu)$ be the rotated item, $y_{i,u}$ its u-th subvector, and $C = \{c_{u,j}\}$ the codebooks, with OPQ values C_0. An item is encoded by the nearest codeword in each subspace and decoded as the concatenation $z_i(C)$. Distances in the rotated basis equal distances in the original space.

Objective. LAQ minimizes

$$J(C) = \frac{\mathcal{L}_{\text{rec}}(C)}{\mathcal{L}_{\text{rec}}(C_0)} + \lambda_{\text{mid}}\,\mathcal{L}_{\text{mid}}(C) + \lambda_{\text{far}}\,\mathcal{L}_{\text{far}}(C), \quad (8)$$

where $\mathcal{L}_{\text{rec}} = \frac{1}{n}\sum_i \|y_i - z_i\|^2$ is the reconstruction error and $\mathcal{L}_{\text{mid}}$ and $\mathcal{L}_{\text{far}}$ are hinge losses at two groups of scales, with weights λ_{mid} and λ_{far}. For a scale group $\mathcal{S}$ and a batch of sampled triplets (i, p, q),

$$\mathcal{L}_{\mathcal{S}} = \frac{1}{s_{\mathcal{S}}}\,\text{mean}\big[\eta\Delta_X - \Delta_Z\big]_+, \quad (9)$$

with Δ_X and Δ_Z the gaps of Corollary 1. The normalizer $s_{\mathcal{S}}$ is the median reference gap of the group, computed once at the start so that both groups contribute on one scale. The middle group draws $k/n \in \{1\%, 5\%\}$. The far group draws k uniformly from $[0.3N, 0.5N]$, the G band. The normalized reconstruction term holds the near end, which product quantization serves well.

Sampling cross pairs. Each step draws 1,024 anchors per group, and each anchor a scale k and a cross pair. Half of the pairs are drawn uniformly: p among the k nearest items of i and q among the other $N - k$. By Corollary 1 these pairs give an unbiased estimate of $\bar{H}_k$, which bounds the violation rate and, through Theorem 1, $1 - \nu_k$. The other half straddle rank k: p is drawn from ranks $k - w$ to $k - 1$ and q from ranks k to $k + w - 1$, with $w = \max(3, \lfloor 0.1k \rfloor)$. Lost and gained neighbors arise at this boundary, as the proof of Theorem 1 shows, so these pairs place half of the gradient where the neighborhood changes.

Straight-through codes. Encoding is a hard assignment. LAQ uses it in the forward pass and the soft assignment $\text{softmax}(-\|y_{i,u} - c_{u,j}\|^2/T)$ in the backward pass [17], so every step encodes all training items with the current codebooks. The temperature falls as $T_t = T_0 \cdot 0.05^{t/t_{\max}}$ over $t_{\max}$ steps, with T_0 the mean squared distance from a subvector to its nearest codeword at the start. The codebooks are updated by Adam with a cosine-decayed rate equal to 0.002 times the standard deviation of the entries of C_0.

Checkpoint on a monitor set. LAQ draws 2,000 training items as a monitor set. Every 250 steps it computes, on the monitor set, ν_k at $k = 0.4$ of the set, the middle of the G band, and the reconstruction error relative to the start. It keeps the codebooks with the highest ν_k among checkpoints whose relative reconstruction error is at most 1.15, and it returns C_0 if no checkpoint exceeds the start. The monitor set is part of the training memory, so held-out items are never consulted.

What LAQ changes. LAQ changes the codebook values and nothing else. The PCA basis, the rotation, the number of bytes per item, and search by asymmetric distance computation (ADC) are those of OPQ. An LAQ index therefore serves queries with the code of an OPQ index and the same latency. One recipe serves every memory and encoder: $t_{\max} = 1{,}000$ steps, $\lambda_{\text{mid}} = \lambda_{\text{far}} = 1$, and $\eta = 1$ (Appendix A). Training holds the $n \times n$ distance matrix of the training memory and its ranks on one GPU. Training memories of 7,336 to 7,672 items fit on one 48 GB GPU.

5 Experimental Setup

Memories. We use five memories (Table 1), three agent memories and two topic corpora. *FinQA* [42] holds pages of annual reports of S&P 500 companies with numerical questions annotated with their supporting facts. We keep the 106 companies in the current index and up to three report pages each. Every sentence and table row is one item, prefixed by the company name and fiscal year. This gives 8,356 items in 11 sectors of the Global Industry Classification Standard (GICS) and 863 questions. *Multi-Doc-2025* [43] is a financial question-answering benchmark over 179 10-K filings of 89 S&P 500 companies for fiscal years 2022 to 2024. Each paragraph and table row is a candidate item with the same prefix. Every filing keeps the passages that hold its questions' evidence plus a random sample, up to 56 passages. This gives 10,024 items and 898 questions with

TABLE 1
The five memories.

Memory	Items	Questions	Structure used
FinQA	8,356	863	company, sector, fiscal year
Multi-Doc-2025	10,024	898	company, sector, fiscal year
LoCoMo	5,882	1,536	conversation, session
20 Newsgroups	10,000	–	topic, topic family
AG News	10,000	–	topic

located evidence. *LoCoMo* [37] is a benchmark of long-term conversational memory for agents, with ten conversations of 19 to 32 sessions, 5,882 turns, and 1,536 answerable questions annotated with their evidence turns. Each turn, prefixed by its speaker, is one item. *20 Newsgroups* [44] and *AG News* [45] contribute 10,000 random documents each, with 20 and 4 topics.

Encoders. Three encoders are retrieval-tuned sentence encoders, used with their query instructions: BGE-base-en-v1.5 and BGE-large-en-v1.5 [46], with 768 and 1,024 dimensions, and E5-base-v2 [47], with 768. Three are encoders of other kinds: the text encoder of MobileCLIP2-L/14 [48] (768 dimensions), RoBERTa-base [49] with mean pooling (768), and averaged GloVe vectors [50] (300). The seventh is the commercial Amazon Titan Text Embeddings V2 [5] at 1,024 dimensions, used on Multi-Doc-2025. All vectors have unit length.

Compressions. There are 18 measurement runs: six encoders on the two agent memories, and the three encoders of other kinds on the two topic corpora. Each run applies 27 compressions fitted on the memory itself. Scalar codes are half precision and 8- and 4-bit codes. Binary codes are sign codes, the same codes with full-precision rescoring of the 40 nearest, and 2- and 4-bit thermometer codes, which set one bit per threshold passed on each coordinate. Truncation, PCA, and Gaussian random projection reduce to $d/2$, $d/4$, $d/8$, and $d/16$ dimensions. PCA to $d/4$ and $d/8$ is also stored with 8-bit codes, and PCA to d and $d/2$ is followed by a random rotation and binarization. Product quantization uses $d/8$, $d/16$, and $d/32$ bytes, and UMAP [51] maps to 16 dimensions. OPQ with and without rescoring is added at $d/8$ bytes. Binary codes are compared by Hamming distance and all others by Euclidean distance on the decoded vectors.

Operations. Each operation runs on the compressed and on the uncompressed memory, and we report their agreement.

- *Retrieval.* Each question retrieves its 10 nearest items (within its conversation for LoCoMo). Agreement is the overlap of the two top-10 lists. Quality is evidence recall@10, the share of a question's evidence among them.
- *Consolidation into themes.* k-means on the compressed memory, started from the centroids of the uncompressed partition, with 11 themes for the financial memories and 20 and 4 for the topic corpora. Agreement is the adjusted Rand index (ARI), which we call theme agreement.
- *Deletion.* For each session or topic, the memory selects as many items as the session or topic holds, those nearest its centroid. This is how a memory selects deletion candidates by meaning when metadata are incomplete. Agreement is the Jaccard overlap of the deleted sets.
- *Company operations.* The information barrier walls off a company by applying deletion to it. Peer groups are each company's five nearest companies by centroid. The company map is the rank correlation of the distances among company centroids with the uncompressed ones; the sector and session maps are defined in the same way.
- *Global references.* These use the subsamples of G. *All distances* is the Spearman correlation of 200,000 pairwise distances [35]. *Triplets* is the accuracy of random triplets [36]. The *median split* is $\nu_{k^\star}$ of Proposition 2.

Across the 27 compressions of a run, we relate each operation's agreement to L and G. We use Spearman correlation and the partial rank correlation with one fidelity given the other.

Held-out protocol for LAQ. LAQ is trained and tested on disjoint companies or conversations. For FinQA and Multi-Doc-2025 the split holds out a quarter of the companies (26 of 106 and 22 of 89); for LoCoMo it holds out two of the ten conversations. Three splits, seeds 2026 to 2028, were fixed in advance. A fourth, seed 2029, was drawn after the LAQ recipe was fixed and run once with it unchanged. PCA, OPQ, and LAQ are fitted on the training part only, and every reported number is computed on the held-out part, with its own questions. The three retrieval-tuned encoders run on all three memories and Titan V2 on Multi-Doc-2025, which gives 40 held-out runs, each comparing LAQ with its own OPQ start at $d/8$ bytes. The other quantizers and the agents of the Supplementary Material were run on the first three splits. On Multi-Doc-2025, queries stay in full precision and are searched by ADC, as an OPQ index serves them. L, G, and the far-end operations use the stored codes.

Quantizers compared. On the held-out Multi-Doc-2025 memories we compare LAQ, at $d/8$ bytes, with eight quantizers. Five store codes of the same size: PQ [4], RQ [21], LSQ++ [22], PCA to $d/8$ dimensions with 8-bit codes, and RaBitQ [25] with d bits plus its per-vector factors (104 and 136 bytes). Sign and ITQ [24] binary codes are rescored on the 40 nearest. ScaNN uses anisotropic quantization [23]. All are fitted on the training companies. ScaNN is fitted on the held-out memory it indexes and returns neighbors for full-precision queries, so we report its L.

6 What the two ends measure

Each end of the profile predicts its own operations. Retrieval follows L, consolidation and deletion follow G, and compressions of one size can keep very different far ends. These results set the target that LAQ meets in Section 7.

6.1 Compressions separate on the (L, G) plane

Compressions of equal severity keep different ends of a memory. Figure 4 places every compression of the two agent memories on the (L, G) plane for the retrieval-tuned encoders, and Figure 10 in Appendix C does so for the other three. Within a memory the two ends are correlated across compressions (Spearman 0.81 to 0.95 over the 18 runs), because a more severe compression lowers both. The compressions still spread off one curve. With the retrieval-tuned encoders, half precision and 8-bit scalar codes keep both ends at 0.97 or above. PCA, PCA with 8-bit codes, and

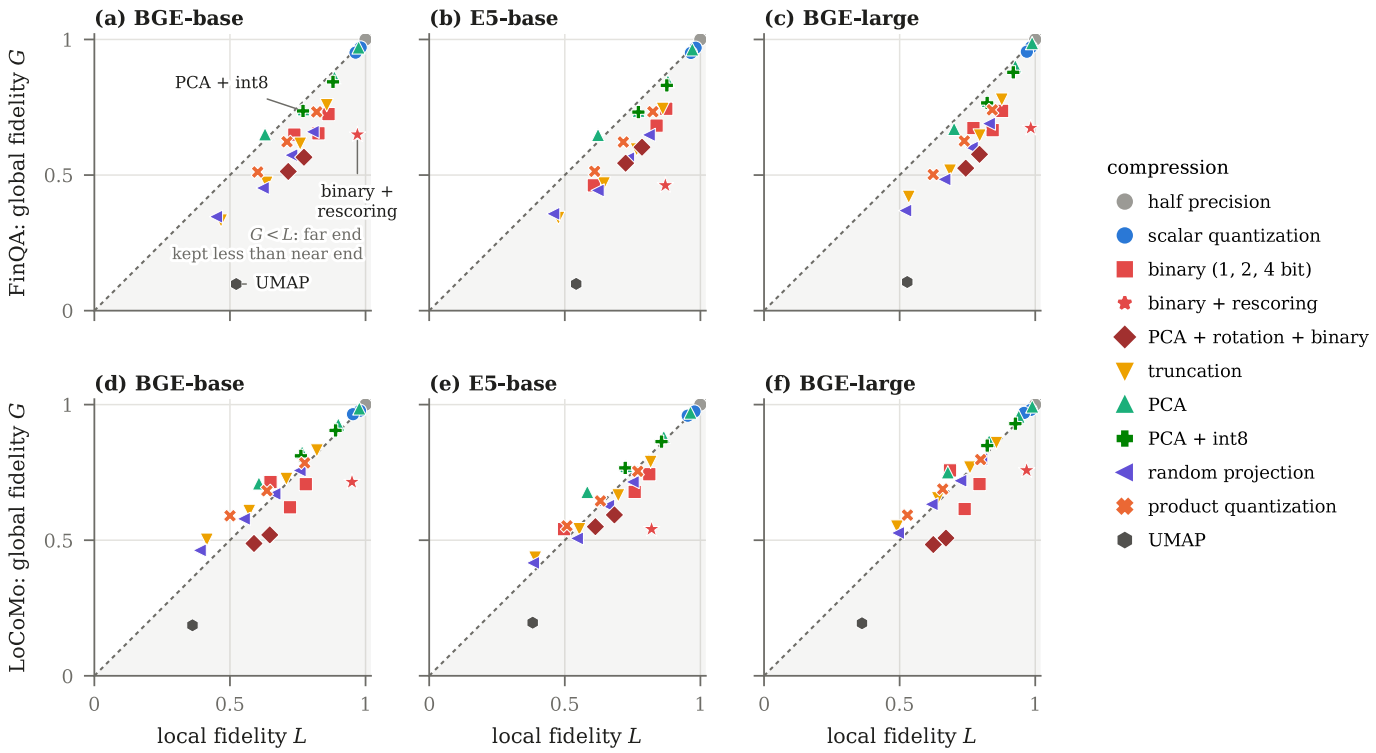


Fig. 4. Compressions of one memory spread off a single curve, and rescored binary codes lie furthest below the diagonal. L and G of 27 compressions of the two agent memories, three retrieval-tuned encoders; Figure 10 gives the other three. Dashed line: $G = L$; shaded: less of the far end kept than of the near end.

product quantization keep G within 0.12 of L. Binarization after PCA and a random rotation places G 0.06 to 0.22 below L. Of the linear compressions, rescored binary codes lie furthest below the diagonal, by 0.21 to 0.41. Rescoring raises L and leaves the stored codes as they were. UMAP to 16 dimensions keeps L at 0.36 to 0.54 and G at 0.10 to 0.20, and other nonlinear maps keep G at 0.22 or less (Table 8). With the encoders of other kinds, PCA and product quantization lie on or above the diagonal, on the agent memories and on the topic corpora (Figures 10 and 11a–f). One-bit codes of GloVe and RoBERTa vectors reach G of 0.58 to 0.81 at L of 0.26 to 0.51. These encoders have coordinates of very unequal variance, and the sign of a low-variance coordinate moves the nearest neighbors more than the nearer half.

6.2 Each end predicts its own operations

Figure 5 plots each operation against both fidelities. Retrieval agreement lies close to one increasing curve in L, with a median within-run Spearman correlation of 0.99 against 0.89 for G. The far-end operations lie closer to a curve in G, with 0.95 against 0.87 for L. L and G move together under a more severe compression, so partial correlations separate the two ends more sharply.

Figure 6 gathers every operation of every run. Retrieval sits at the lower right. Its partial correlation with L given G is 0.83 to 0.99 (Tables 5 and 6 in Appendix C). Consolidation, deletion, and the session and topic maps sit at the upper left. With the retrieval-tuned encoders, consolidation of the financial memory into themes follows G. Its partial correlations with G given L are 0.68, 0.76, and 0.66 for BGE-base, E5-base, and BGE-large, and with L given G they are 0.04, 0.01, and -0.02. Deleting a LoCoMo session has 0.86, 0.55, and 0.93 with G given L, and -0.11, 0.36, and -0.14 with L given G. Session and topic maps follow G at 0.56 to 0.93 given L, and the two global references, all distances and triplets, at 0.78 to 1.00. The operations defined on companies sit between the two ends. Peer groups, the barrier, and the company and sector maps compare 106 company centroids. Each is thus a nearest-neighbor question in the space of companies. With the retrieval-tuned encoders they follow L (0.39 to 0.65 given G) at least as closely as G (0.21 to 0.51 given L). The rule is one of scale: an operation follows the end of the profile at which it compares items.

6.3 Rescoring repairs the near end and keeps the stored codes

Rescoring repairs the near end, and the far end stays with the stored codes. Disk-based vector search keeps binary codes in memory and reranks candidates with full-precision vectors read from disk. Figure 7 reproduces this with the 40 nearest codes. On FinQA, rescoring raises L from 0.74 to 0.97 with BGE-base, from 0.61 to 0.87 with E5-base, and from 0.77 to 0.98 with BGE-large. It raises retrieval agreement from 0.57 to 0.87, 0.30 to 0.53, and 0.62 to 0.90. G stays at the value of the binary codes, and so does every operation computed on them. The financial memory consolidates into themes at 0.79, 0.63, and 0.82. A LoCoMo session is deleted with agreement 0.56, 0.35, and 0.57. By the recall check, rescored binary codes are close to the original memory. By G, they keep the far end of plain binary codes. Rescoring works at query time, and the far end is a property of the stored codes. The same

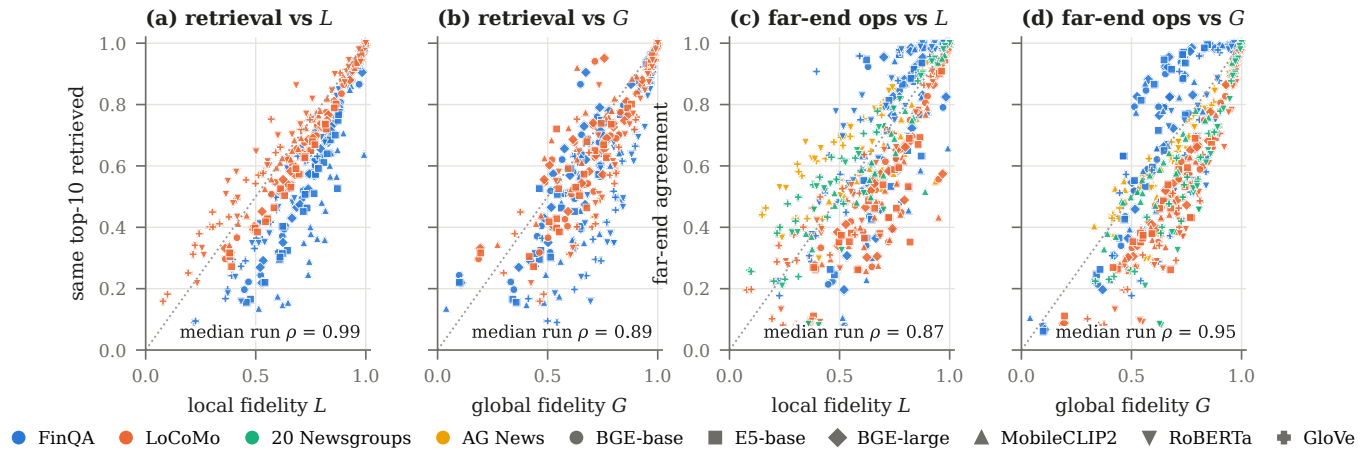


Fig. 5. Retrieval follows L, and the far-end operations follow G. Agreement with the uncompressed memory, every compression and run. (a, b) Retrieval on the two agent memories. (c, d) Consolidation into themes for FinQA, deletion for LoCoMo and the topic corpora. ρ: median over runs of the Spearman correlation across the 27 compressions of a run.

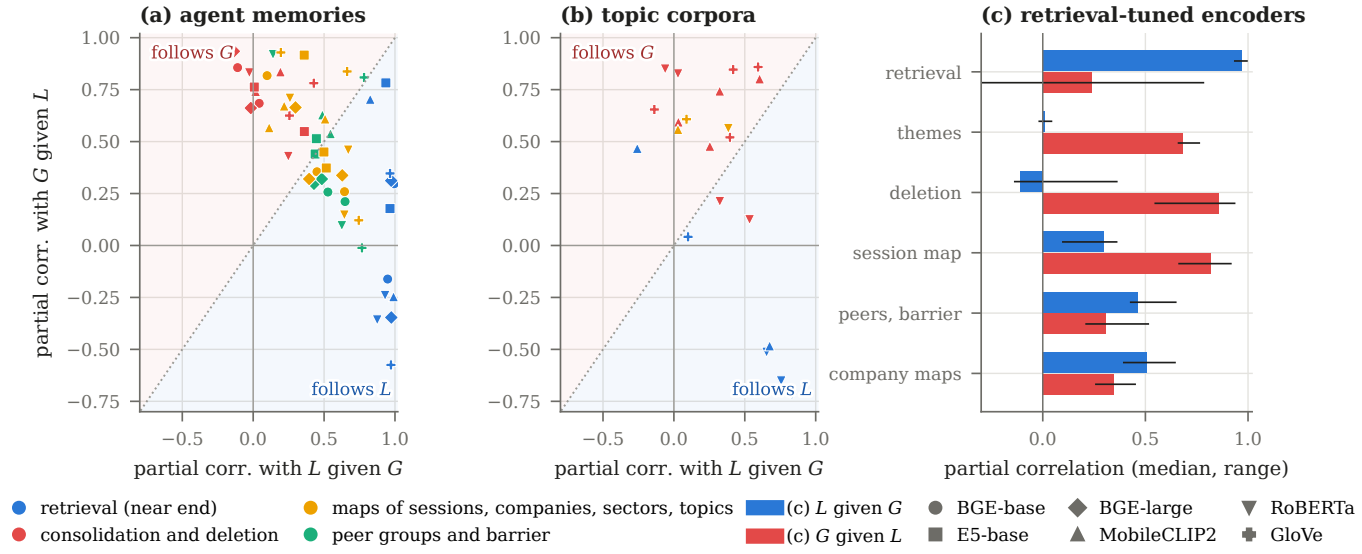


Fig. 6. Each operation follows the end at which it compares items: retrieval lies in the L region, consolidation and deletion in the G region. Partial rank correlation with G given L against that with L given G, across the 27 compressions of each run. (a) Agent memories, (b) topic corpora; color: operation; shape: encoder. (c) Medians and ranges over the six runs of the retrieval-tuned encoders.

holds for every code family that keeps Euclidean distances (Table 7).

With the retrieval-tuned encoders, evidence recall follows L, with partial correlations of 0.71 to 0.83 given G on FinQA and 0.61 to 0.87 on LoCoMo (Figure 7c). Financial text repeats across years, so a retrieved passage can come from the right company and the wrong fiscal year. The uncompressed memories of the retrieval-tuned encoders take 0.18 of their right-company passages from another year. Every linear compression keeps this share between 0.15 and 0.24 (Figure 7d).

6.4 Same bytes, different ends

Memories of the same size differ most at the far end. At $d/8$ bytes per item, 32 times smaller than single precision, we compare seven compressions. A memory can hold one-bit codes with or without rescoring, rotated one-bit codes, product-quantization codes, $d/8$ principal components in 8 bits, or OPQ codes with or without rescoring (Figure 8; Table 9 in Appendix C lists all 27 compressions of FinQA with BGE-base). On FinQA with BGE-base, rescored binary codes have the highest L of the first five, 0.97, and consolidate the memory into themes at 0.79. PCA with 8-bit codes keeps the themes at 0.96. The same order holds with E5-base (0.96 against 0.63) and with BGE-large at 128 bytes (0.97 against 0.82). On LoCoMo, product quantization and PCA with 8-bit codes delete a session with agreement 0.43 to 0.68, and binary codes with 0.35 to 0.57. OPQ has the highest G of the seven with every encoder on both memories: 0.82 to 0.86 on FinQA and 0.85 to 0.92 on LoCoMo. It deletes a LoCoMo session with agreement 0.70 to 0.82. Rescored, it returns 0.99 to 1.00 of the same top-10 memories. OPQ thus keeps both ends best among the seven compressions at this size, and it is the starting point of LAQ.

Company-level results show where a compression moves themes. With rescored binary BGE-base codes, the passages

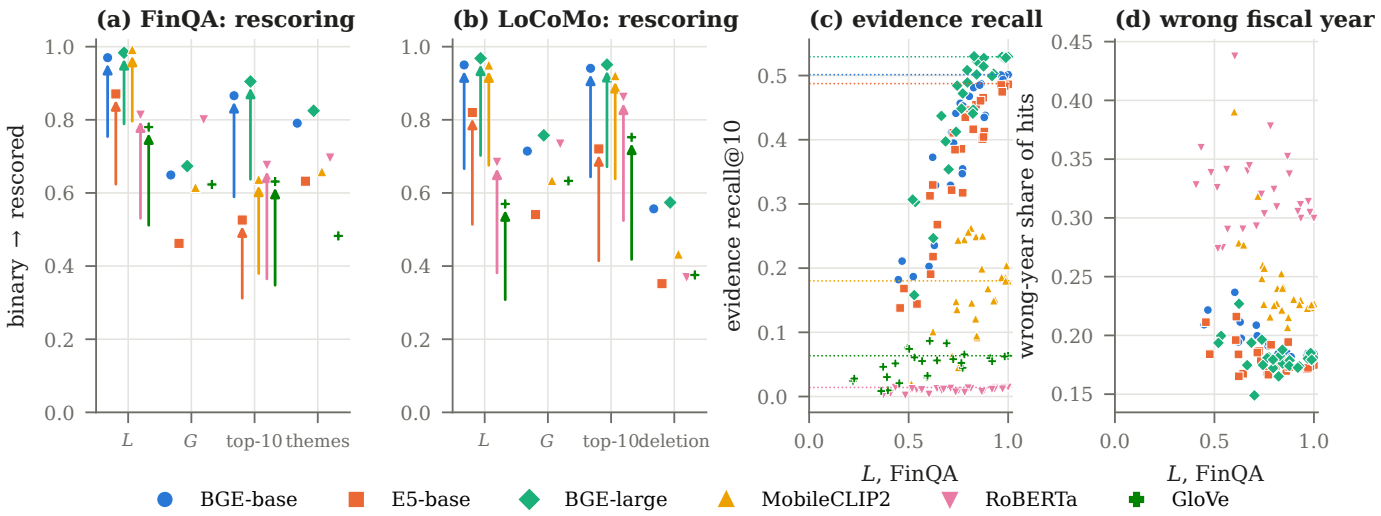


Fig. 7. Rescoring raises L and retrieval and leaves G and the far-end operations at the values of the binary codes. (a, b) Binary codes before and after full-precision rescoring of the 40 nearest codes; arrows run from binary to rescored, and a marker without an arrow is unchanged. (c) Evidence recall@10 on FinQA rises with L; dotted lines: uncompressed memories. (d) The share of right-company passages from another fiscal year stays near its uncompressed level.

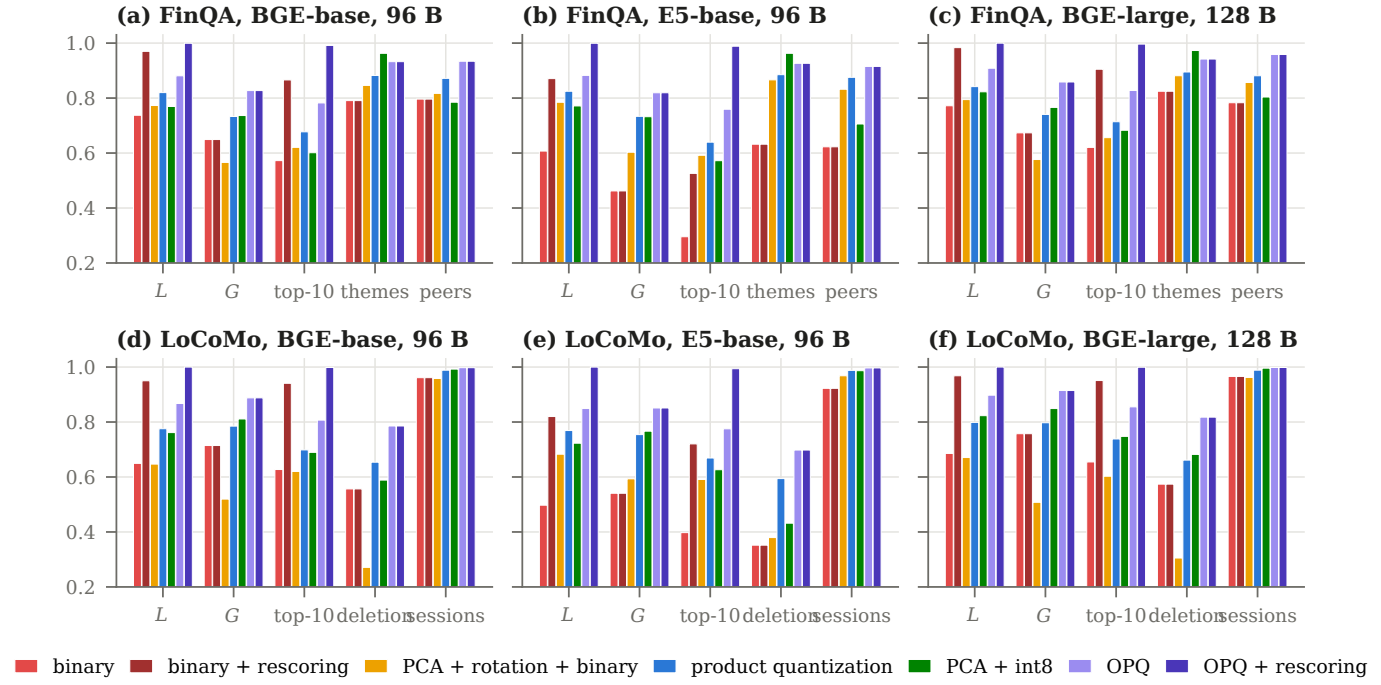


Fig. 8. At $d/8$ bytes, OPQ keeps the highest G of seven compressions, and rescored binary codes keep a high L with a lower G. Fidelity and agreement with the uncompressed memory, retrieval-tuned encoders.

of every FinQA company keep 91 to 100 percent of their 10 nearest neighbors. Still, 5 of the 106 companies have 20 percent or more of their passages consolidated into a different theme (Figure 11g). With PCA and 8-bit codes, every company keeps at least 86 percent of its passages in their theme.

7 LAQ ON HELD-OUT MEMORIES

LAQ raises the far end of every held-out memory, keeps the near end, and lifts the operations that follow G. It also has the highest G among quantizers of the same size.

7.1 LAQ raises the far end of every held-out store

LAQ raises G in all 40 held-out runs (Table 2, Figure 9e). With the retrieval-tuned encoders the gain is 0.005 to 0.017. Every gain exceeds the standard deviation of G over subsamples, which is at most 0.0023. With Titan V2, whose OPQ codes keep the least of the far end, the gain is 0.032 to 0.038. L rises in 34 of the 40 runs and stays within 0.0023 of OPQ in all of them, so the far end is gained with the near end kept.

The gain covers the whole profile (Figure 9). On the 16 held-out Multi-Doc-2025 memories, LAQ lifts ν_k above OPQ across the scales from $k = 10$ to $n/2$. The gain grows toward

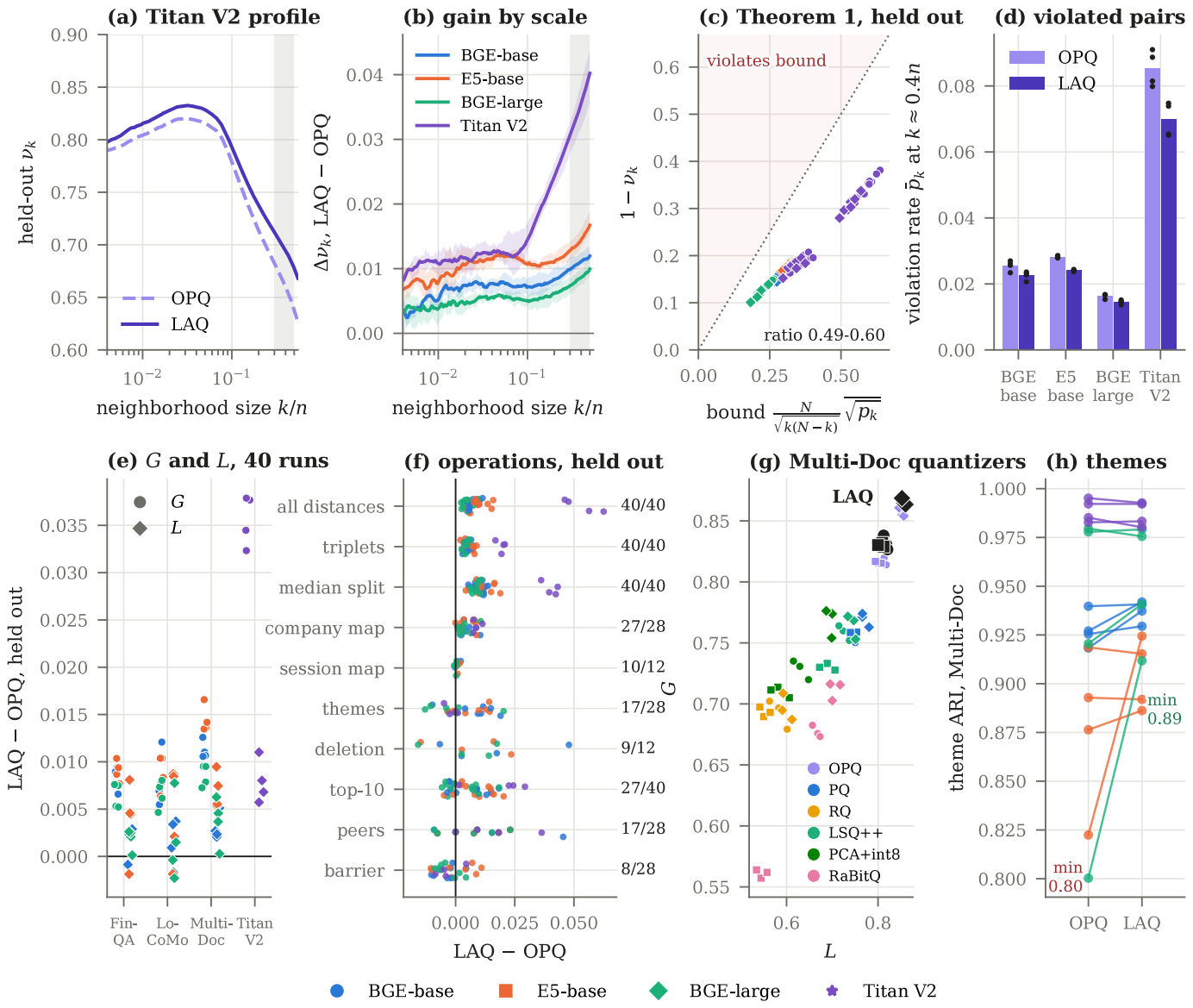


Fig. 9. LAQ raises the far end of every held-out memory by lowering rank violations, and the gain reaches the far-end operations. Top, held-out Multi-Doc-2025: (a) profiles of OPQ and LAQ for Titan V2, mean of four splits; gray: the G band. (b) Gain in ν_k from $k = 10$ to $n/2$; bands: ranges over four splits. (c) The bound of Theorem 1 holds at every point (OPQ circles, LAQ diamonds). (d) Violation rate $\bar{p}_k$ at $k \approx 0.4n$; dots: splits. Bottom, all held-out runs: (e) change in G and L, positive for G in all 40 runs. (f) Change in each operation's agreement; right: runs in which LAQ is higher. (g) G and L of LAQ (black) and six other quantizers at $d/8$ bytes, Multi-Doc-2025, first three splits. (h) Theme agreement on Multi-Doc-2025; the lowest value rises from 0.80 to 0.89.

the G band, where LAQ trains. Theorem 1 holds at all 160 combinations of memory, quantizer, and scale, with $1 - \nu_k$ at 0.49 to 0.60 of the bound. The theorem also explains the mechanism. At $k \approx 0.4n$, LAQ lowers the violation rate $\bar{p}_k$ by 10 to 16 percent with the retrieval-tuned encoders and by 16 to 20 percent with Titan V2, and through the bound this lowers $1 - \nu_k$ in the G band.

7.2 The gain reaches the operations that follow G

The operations that follow G in Section 6 move with it (Figure 9e, f). The two global references and the median split rise in all 40 runs. The correlation of all distances rises by 0.002 to 0.016 with the retrieval-tuned encoders and by 0.046 to 0.063 with Titan V2. The median split rises by 0.004 to 0.019 and by 0.036 to 0.043. The company map rises in 27 of 28 runs and the session map in 10 of 12, with the other two within 0.001.

Consolidation into themes gains most where OPQ consolidates least. On Multi-Doc-2025 with the retrieval-tuned encoders, the mean theme agreement rises from 0.908 to 0.931. The lowest value over the 12 runs rises from 0.800 to 0.886 (Figure 9h). With Titan V2, both keep the themes at 0.98 or above. On FinQA the mean rises from 0.911 to 0.914. Deleting a LoCoMo session gains in 9 of 12 runs, with mean agreement rising from 0.686 to 0.695.

Retrieval is served by the near end, which LAQ keeps. Top-10 agreement rises in 27 of 40 runs. Mean evidence recall@10 stays within 0.005 of OPQ in every memory: 0.545 and 0.546 on FinQA, 0.649 and 0.649 on LoCoMo, 0.795 and 0.796 on Multi-Doc-2025, and 0.749 and 0.754 with Titan V2.

TABLE 2
LAQ raises G over its OPQ start in every held-out run. $d/8$ bytes; runs in which LAQ is higher, means over runs, and the range of the change. Retrieval-tuned encoders (BGE-base, E5-base, BGE-large), four splits each.

Memory		higher	OPQ	LAQ	LAQ − OPQ
FinQA	G	12/12	.800	.808	$+.005$ to $+.010$
	L	10/12	.861	.864	$-.002$ to $+.008$
LoCoMo	G	12/12	.845	.853	$+.005$ to $+.012$
	L	8/12	.820	.822	$-.002$ to $+.009$
Multi-Doc	G	12/12	.830	.842	$+.007$ to $+.017$
	L	12/12	.823	.828	$+.0003$ to $+.010$
Multi-Doc, Titan V2	G	4/4	.656	.692	$+.032$ to $+.038$
	L	4/4	.790	.798	$+.006$ to $+.011$

TABLE 3
LAQ has the highest G with every encoder. G / L on held-out Multi-Doc-2025 at $d/8$ bytes, means over the first three splits. RaBitQ uses 104 and 136 bytes. ScaNN is fitted on the held-out memory and searched with full-precision queries; we report its L.

	BGE-base	E5-base	BGE-large	Titan V2
LAQ	**.832** / **.817**	**.830** / **.807**	**.866** / **.857**	**.696** / .800
OPQ	.820 / .814	.816 / .801	.857 / .852	.659 / .792
PQ	.754 / .743	.759 / .749	.769 / .771	.661 / .787
LSQ++	.759 / .725	.730 / .689	.764 / .744	.543 / .679
RQ	.693 / .582	.693 / .552	.697 / .598	.429 / .511
PCA + int8	.729 / .631	.710 / .584	.768 / .695	.521 / .578
RaBitQ	.677 / .666	.561 / .545	.712 / .704	.656 / .776
ScaNN	– / .615	– / .507	– / .578	– / **.837**

Peer groups and the information barrier are nearest-neighbor questions among company centroids, and Section 6 found that they follow L. LAQ raises peer agreement in 17 of 28 runs and keeps barrier agreement within 0.011 of OPQ in every run. Every run is listed in Appendix B, and the answers of two agents over these memories are in Section S1 of the Supplementary Material.

7.3 LAQ against other quantizers at the same size

LAQ leads the quantizers of its size. Table 3 and Figure 9g compare LAQ with OPQ and six other quantizers on the held-out Multi-Doc-2025 memories of the first three splits. With each retrieval-tuned encoder, LAQ has the highest G and the highest L of all codes without rescoring in each of the nine runs. LSQ++, the strongest of the additive quantizers here, reaches G of 0.73 to 0.77, against 0.83 to 0.87 for LAQ. With Titan V2, LAQ has the highest G in each of the three runs, and ScaNN, fitted on the held-out memory itself, has the highest L. With rescoring of the 40 nearest codes, LAQ reaches an L of 0.995 to 0.999 in all twelve runs. This is above rescored sign codes (0.861 to 0.996) and rescored ITQ codes (0.938 to 0.994) in each run, and the stored LAQ codes keep the highest G.

8 Discussion

Why the far end can be learned. Ordering near–far pairs controls the far end tightly, and this is what makes LAQ work. The bound has a factor near 2 across the band of G and near $\sqrt{N/10}$ at $k = 10$. Reconstruction and nearest-neighbor objectives serve the near end. LAQ uses the two facts together: a hinge at middle and far scales, a reconstruction term for the near end, and a checkpoint chosen on the far end of a monitor set. The gain is largest where the OPQ codes keep least of the far end, as with Titan V2. It reaches the operations that follow G, and the lowest theme agreement on Multi-Doc-2025 rises from 0.80 to 0.89.

What to report. A compressed memory can ship with (L, G) against the memory it replaces, computed on the operator's own vectors. L says whether a query returns the same items. G says whether the memory keeps its shape: which items are consolidated together, which are deleted together, and how sessions, companies, and themes relate. Both take seconds, with G from 2,000 items, and neither needs labels. Where a memory serves consolidation or deletion, G belongs in its acceptance test, and the compression is chosen on the (L, G) frontier for the bytes available. Theorem 1 adds a diagnostic: the violation rate of cross pairs at a scale shows which part of the profile a compression loses.

Financial memory. The financial memories carry the clearest stakes. A consolidation that groups filings differently changes the sector briefs an analyst agent writes. A deletion that selects different passages changes what a client request removes. Peer groups, the information barrier, and the company map compare company centroids. A check of these operations combines L with agreement measured on the firm's own companies.

Scope and extensions. Agreement is measured against the uncompressed memory, the reference for certification, and quality is reported beside it where ground truth exists. One LAQ recipe serves every memory and encoder. It was fixed on the first three splits and confirmed on a fourth, and every held-out company and conversation was unseen when PCA, OPQ, and LAQ were fitted. Residual and additive codebooks are natural next targets for the same objective, and sampling the distance matrix would extend training to larger memories.

9 Conclusion

Agent memories are compressed and accepted by their nearest neighbors, and agents also consolidate, delete, and map what they hold. One label-free profile measures both ends. Its near end predicts retrieval and its far end predicts consolidation and deletion, each beyond the other. Our main result is that the order of near and far items bounds the profile at every scale, and that this bound can be trained. Locality-aware quantization learns that order and raises the global fidelity of every held-out memory at the same bytes and search as the OPQ index it replaces. For agent memory, and for any vector index that serves clustering, deduplication, or maps beside search, (L, G) gives an acceptance test for both ends and LAQ gives a way to pass it.

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Appendix A
LAQ recipe

Every LAQ run uses one recipe: $M = d/8$ subspaces of 8 dimensions with 256 codewords each, the full PCA basis, 1,000 steps, 1,024 anchors per scale group and step, $\lambda_{\text{mid}} = \lambda_{\text{far}} = 1$, margin $\eta = 1$, Adam with a learning rate of 0.002 times the standard deviation of C_0 and cosine decay, temperature from T_0 to $0.05\,T_0$, a monitor set of 2,000 training items checked every 250 steps, and a reconstruction limit of 1.15 for a checkpoint. The scale normalizers s_{mid} and s_{far} are medians of 16,384 reference gaps sampled at the start.

Appendix B
LAQ against OPQ by run

Appendix C
Additional measurement results

The topic corpora repeat the pattern of the agent memories, and the tables give every number behind the main figures. Figure 10 places the 27 compressions of the two agent memories on the (L, G) plane for the three encoders of other kinds.

Figure 11 places the 27 compressions of the two topic corpora on the (L, G) plane and breaks the 96-byte comparison of FinQA down by company. Tables 5–9 give the correlations behind Figures 5 and 6, rescoring for every code family that keeps L_2 distances, nonlinear maps, and every compression of the financial memory.

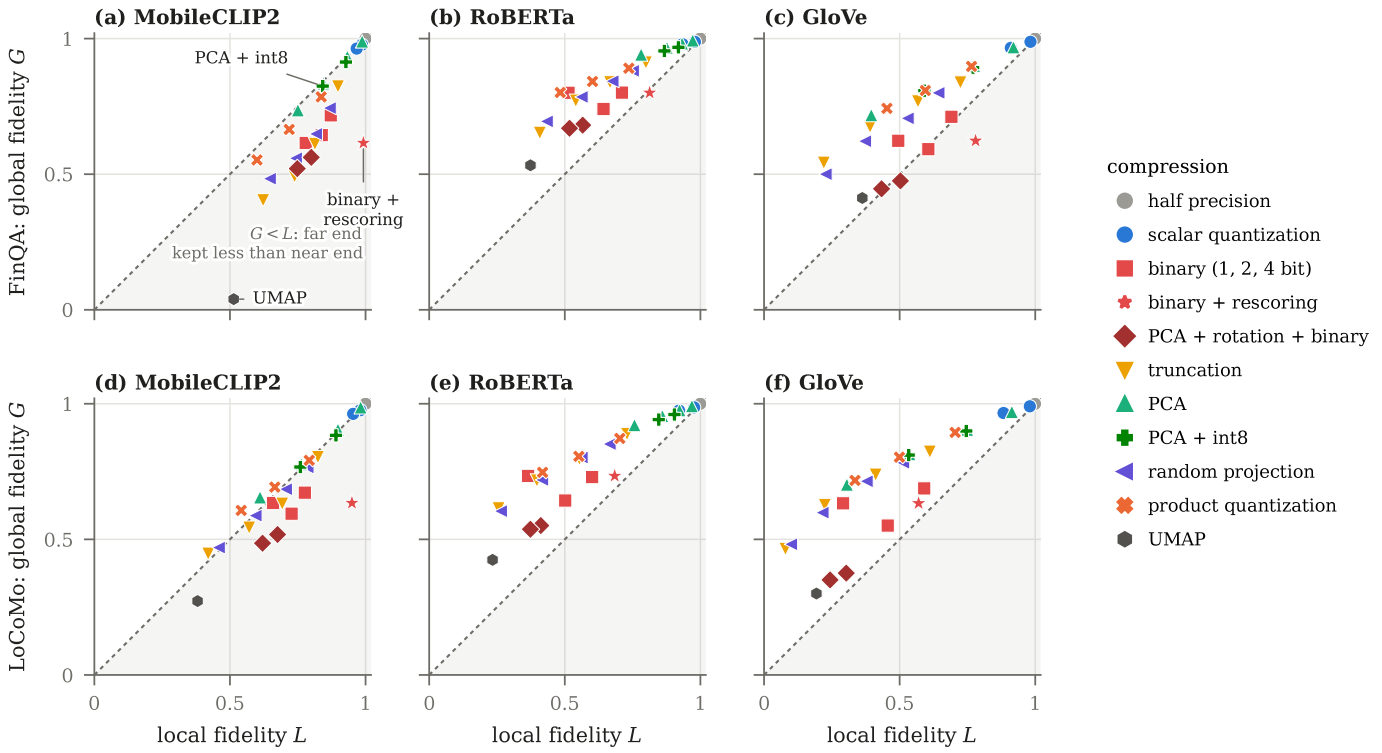


Fig. 10. With encoders of other kinds, PCA and product quantization lie on or above the diagonal, and one-bit codes of GloVe and RoBERTa keep more of the far end than of the near end. L and G of 27 compressions of the two agent memories; dashed line: $G = L$.

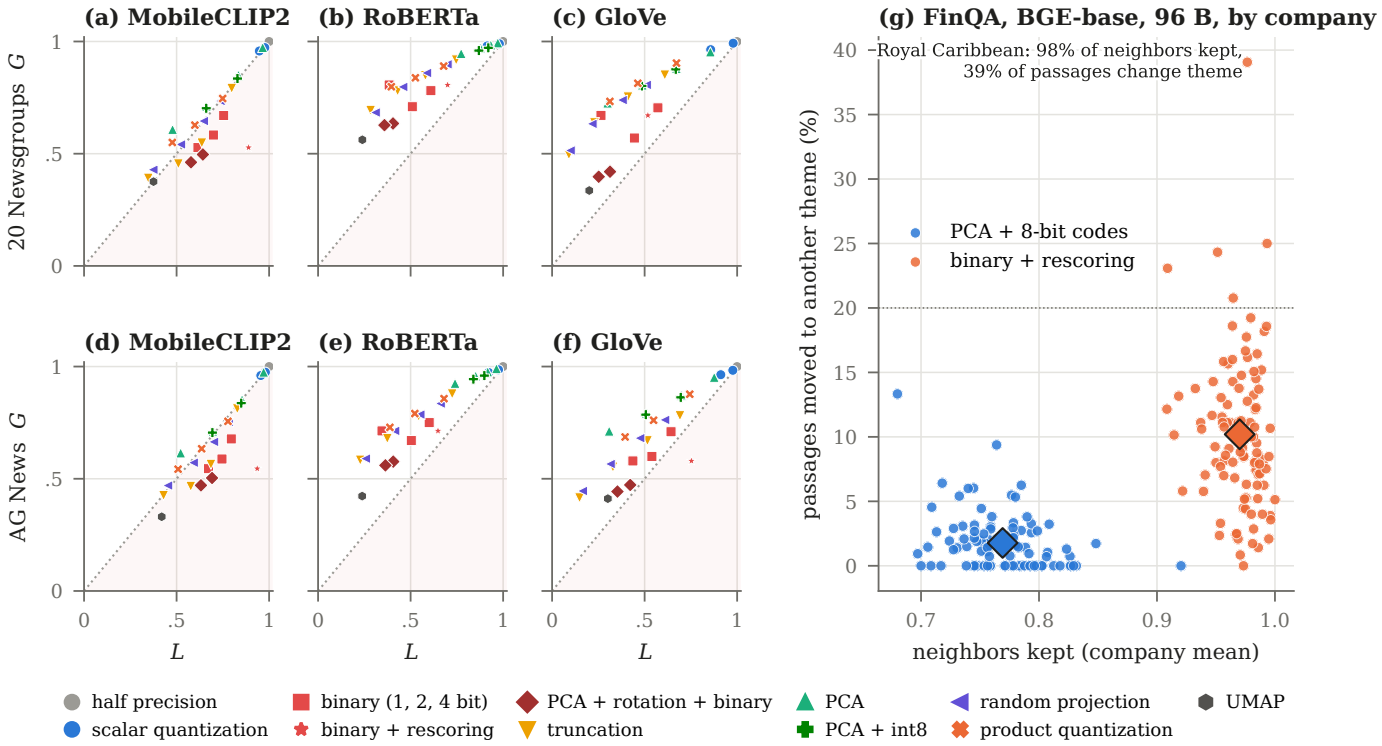


Fig. 11. On the topic corpora, PCA and product quantization lie on or above the diagonal (a–f), and rescored binary codes keep neighbors yet move up to 39 percent of a company's passages to another theme (g). (a–f) L and G of 27 compressions, three encoders of other kinds; dotted line: $G = L$. (g) The two 96-byte compressions of FinQA (BGE-base) by company; diamonds: means; dotted line: 20 percent moved.

TABLE 4
LAQ raises G over OPQ on every held-out memory. G and L of OPQ and LAQ, four splits.

Memory	Encoder	Split	G OPQ	G LAQ	L OPQ	L LAQ
FinQA	BGE-base	2026	0.7927	0.8021	0.8537	0.8566
		2027	0.7930	0.8020	0.8579	0.8570
		2028	0.7885	0.7951	0.8534	0.8557
		2029	0.7958	0.8035	0.8489	0.8514
	E5-base	2026	0.7872	0.7959	0.8413	0.8458
		2027	0.7858	0.7952	0.8463	0.8544
		2028	0.7781	0.7884	0.8440	0.8486
		2029	0.7867	0.7944	0.8469	0.8450
	BGE-large	2026	0.8227	0.8280	0.8819	0.8840
		2027	0.8209	0.8261	0.8844	0.8869
		2028	0.8185	0.8260	0.8884	0.8910
		2029	0.8266	0.8342	0.8879	0.8880
LoCoMo	BGE-base	2026	0.8667	0.8744	0.8186	0.8224
		2027	0.8572	0.8641	0.8349	0.8358
		2028	0.8319	0.8440	0.8148	0.8182
		2029	0.8722	0.8777	0.8319	0.8302
	E5-base	2026	0.7979	0.8043	0.7651	0.7671
		2027	0.7681	0.7785	0.7859	0.7946
		2028	0.7703	0.7807	0.7612	0.7697
		2029	0.8072	0.8156	0.7825	0.7806
	BGE-large	2026	0.8921	0.8994	0.8496	0.8574
		2027	0.8928	0.8989	0.8813	0.8790
		2028	0.8678	0.8759	0.8522	0.8518
		2029	0.9128	0.9175	0.8587	0.8602
Multi-Doc	BGE-base	2026	0.8196	0.8306	0.8132	0.8183
		2027	0.8140	0.8266	0.8181	0.8208
		2028	0.8274	0.8381	0.8102	0.8122
		2029	0.8165	0.8271	0.8183	0.8207
	E5-base	2026	0.8154	0.8320	0.8018	0.8092
		2027	0.8154	0.8290	0.8068	0.8122
		2028	0.8167	0.8302	0.7944	0.7999
		2029	0.8140	0.8282	0.8037	0.8132
	BGE-large	2026	0.8552	0.8647	0.8526	0.8572
		2027	0.8540	0.8635	0.8564	0.8601
		2028	0.8608	0.8686	0.8462	0.8524
		2029	0.8545	0.8617	0.8557	0.8559
	Titan V2	2026	0.6680	0.7025	0.7942	0.8010
		2027	0.6434	0.6811	0.7962	0.8043
		2028	0.6651	0.7030	0.7848	0.7958
		2029	0.6480	0.6803	0.7852	0.7909

TABLE 5
Retrieval follows L, and consolidation and deletion follow G. Spearman correlation across 27 compressions of each operation's agreement with the uncompressed memory (label precision for the topic corpora) with L and with G, and the partial rank correlation with each end given the other. Agent memories, FinQA.

Memory, encoder	Operation	End	L	G	$L \mid G$	$G \mid L$
FinQA, BGE-base	same top-10 retrieved	near	1.00	0.89	0.99	0.30
	same themes (ARI)	far	0.80	0.90	0.04	0.68
	same peer groups	far	0.87	0.83	0.53	0.26
	information barrier	far	0.91	0.84	0.65	0.21
	company map	far	0.91	0.86	0.64	0.26
	sector map	far	0.86	0.84	0.45	0.35
FinQA, E5-base	same top-10 retrieved	near	0.98	0.94	0.94	0.78
	same themes (ARI)	far	0.83	0.93	0.01	0.76
	same peer groups	far	0.89	0.90	0.45	0.51
	information barrier	far	0.88	0.88	0.44	0.44
	company map	far	0.90	0.89	0.50	0.45
	sector map	far	0.89	0.87	0.52	0.37
FinQA, BGE-large	same top-10 retrieved	near	1.00	0.92	0.97	0.31
	same themes (ARI)	far	0.83	0.91	-0.02	0.66
	same peer groups	far	0.88	0.86	0.43	0.30
	information barrier	far	0.90	0.88	0.48	0.32
	company map	far	0.94	0.91	0.63	0.34
	sector map	far	0.87	0.86	0.39	0.32
FinQA, MobileCLIP2	same top-10 retrieved	near	0.95	0.92	0.83	0.70
	same themes (ARI)	far	0.77	0.90	0.02	0.74
	same peer groups	far	0.88	0.90	0.49	0.63
	information barrier	far	0.88	0.88	0.55	0.54
	company map	far	0.88	0.90	0.51	0.61
	sector map	far	0.73	0.83	0.11	0.57
FinQA, RoBERTa	same top-10 retrieved	near	0.98	0.88	0.93	-0.24
	same themes (ARI)	far	0.84	0.86	0.25	0.43
	same peer groups	far	0.91	0.85	0.63	0.10
	information barrier	far	0.91	0.99	0.14	0.92
	company map	far	0.95	0.93	0.67	0.46
	sector map	far	0.92	0.87	0.64	0.15
FinQA, GloVe	same top-10 retrieved	near	0.99	0.83	0.97	0.35
	same themes (ARI)	far	0.77	0.86	0.26	0.63
	same peer groups	far	0.90	0.72	0.77	-0.01
	information barrier	far	0.92	0.93	0.78	0.81
	company map	far	0.89	0.94	0.66	0.84
	sector map	far	0.90	0.76	0.74	0.12

TABLE 6
Continuation of Table 5: LoCoMo and the topic corpora.

Memory, encoder	Operation	End	L	G	$L \mid G$	$G \mid L$
LoCoMo, BGE-base	same top-10 retrieved	near	0.99	0.89	0.95	-0.16
	same memories deleted	far	0.87	0.97	-0.11	0.86
	session map	far	0.89	0.97	0.10	0.82
LoCoMo, E5-base	same top-10 retrieved	near	0.99	0.89	0.96	0.18
	same memories deleted	far	0.87	0.90	0.36	0.55
	session map	far	0.90	0.98	0.36	0.92
LoCoMo, BGE-large	same top-10 retrieved	near	0.99	0.90	0.97	-0.35
	same memories deleted	far	0.90	0.99	-0.14	0.93
	session map	far	0.91	0.94	0.30	0.66
LoCoMo, MobileCLIP2	same top-10 retrieved	near	1.00	0.88	0.99	-0.25
	same memories deleted	far	0.88	0.96	0.19	0.84
	session map	far	0.86	0.92	0.22	0.67
LoCoMo, RoBERTa	same top-10 retrieved	near	0.97	0.89	0.87	-0.36
	same memories deleted	far	0.92	0.98	-0.03	0.83
	session map	far	0.93	0.97	0.26	0.71
LoCoMo, GloVe	same top-10 retrieved	near	0.99	0.85	0.97	-0.57
	same memories deleted	far	0.92	0.96	0.43	0.78
	session map	far	0.90	0.99	0.20	0.93
20 Newsgroups, MobileCLIP2	label precision of 10 neighbors	near	0.59	0.29	0.68	-0.48
	same memories deleted	far	0.90	0.95	0.60	0.80
	same themes (ARI)	far	0.77	0.81	0.25	0.48
	topic map	far	0.68	0.79	0.03	0.56
20 Newsgroups, RoBERTa	label precision of 10 neighbors	near	0.62	0.41	0.76	-0.65
	same memories deleted	far	0.92	0.98	-0.06	0.85
	same themes (ARI)	far	0.91	0.88	0.53	0.12
	topic map	far	0.93	0.95	0.38	0.56
20 Newsgroups, GloVe	label precision of 10 neighbors	near	0.67	0.34	0.83	-0.71
	same memories deleted	far	0.93	0.97	0.59	0.86
	same themes (ARI)	far	0.87	0.89	0.40	0.52
	topic map	far	0.80	0.88	0.09	0.61
AG News, MobileCLIP2	label precision of 10 neighbors	near	0.31	0.49	-0.26	0.47
	same memories deleted	far	0.86	0.93	0.32	0.74
	same themes (ARI)	far	0.73	0.83	0.03	0.59
AG News, RoBERTa	label precision of 10 neighbors	near	0.62	0.46	0.65	-0.51
	same memories deleted	far	0.93	0.98	0.03	0.83
	same themes (ARI)	far	0.86	0.85	0.32	0.21
AG News, GloVe	label precision of 10 neighbors	near	0.24	0.22	0.10	0.04
	same memories deleted	far	0.86	0.95	0.42	0.85
	same themes (ARI)	far	0.62	0.80	-0.14	0.65

TABLE 7
Rescoring raises L and retrieval for every code family, and G and the far-end operations keep the values of the stored codes. Code families that keep Euclidean distances, $d/8$ bytes per item. The 40 nearest codes are reranked with the original vectors; L and retrieval come from the reranked lists, G and consolidation or deletion from the stored codes. Retrieval: overlap of the top-10 lists. Last column: ARI of consolidation into themes (FinQA) or agreement of session deletion (LoCoMo).

Memory	Encoder	Compression	L	G	Retrieval	Themes / deletion
FinQA	BGE-base (96 B)	PCA + int8	0.77	0.74	0.60	0.96
		PCA + int8 + rescoring	0.99	0.74	0.91	0.96
		PQ	0.82	0.73	0.68	0.88
		PQ + rescoring	0.99	0.73	0.95	0.88
		OPQ	0.88	0.83	0.78	0.93
		OPQ + rescoring	1.00	0.83	0.99	0.93
FinQA	E5-base (96 B)	PCA + int8	0.77	0.73	0.57	0.96
		PCA + int8 + rescoring	0.99	0.73	0.89	0.96
		PQ	0.82	0.73	0.64	0.89
		PQ + rescoring	1.00	0.73	0.94	0.89
		OPQ	0.88	0.82	0.76	0.93
		OPQ + rescoring	1.00	0.82	0.99	0.93
FinQA	BGE-large (128 B)	PCA + int8	0.82	0.77	0.68	0.97
		PCA + int8 + rescoring	1.00	0.77	0.96	0.97
		PQ	0.84	0.74	0.71	0.89
		PQ + rescoring	1.00	0.74	0.97	0.89
		OPQ	0.91	0.86	0.83	0.94
		OPQ + rescoring	1.00	0.86	1.00	0.94
LoCoMo	BGE-base (96 B)	PCA + int8	0.76	0.81	0.69	0.59
		PCA + int8 + rescoring	1.00	0.81	0.99	0.59
		PQ	0.78	0.79	0.70	0.65
		PQ + rescoring	1.00	0.79	0.98	0.65
		OPQ	0.87	0.89	0.81	0.79
		OPQ + rescoring	1.00	0.89	1.00	0.79
LoCoMo	E5-base (96 B)	PCA + int8	0.72	0.77	0.63	0.43
		PCA + int8 + rescoring	0.99	0.77	0.97	0.43
		PQ	0.77	0.75	0.67	0.59
		PQ + rescoring	0.99	0.75	0.96	0.59
		OPQ	0.85	0.85	0.78	0.70
		OPQ + rescoring	1.00	0.85	0.99	0.70
LoCoMo	BGE-large (128 B)	PCA + int8	0.82	0.85	0.75	0.68
		PCA + int8 + rescoring	1.00	0.85	1.00	0.68
		PQ	0.80	0.80	0.74	0.66
		PQ + rescoring	1.00	0.80	0.98	0.66
		OPQ	0.90	0.91	0.86	0.82
		OPQ + rescoring	1.00	0.91	1.00	0.82

TABLE 8
Nonlinear maps keep G at 0.22 or less. Maps fitted to the memory (BGE-base), kept in single precision. Queries are mapped by the fitted transform (UMAP [51], PHATE [52]) or placed at the inverse-distance-weighted mean of the positions of their 10 nearest stored items (t-SNE [11], SR-tSNE [53]). Retrieval: overlap of the top-10 lists. Last column: ARI of consolidation into themes (FinQA) or agreement of session deletion (LoCoMo).

Memory	Map	Bytes	L	G	Retrieval	Themes / deletion
FinQA	UMAP, 16 dims	64	0.523	0.10	0.24	0.08
	PCA to 50, UMAP to 16	64	0.474	0.18	0.16	0.21
	PHATE, 16 dims	64	0.537	0.15	0.19	0.08
	UMAP, 2 dims	8	0.453	0.07	0.18	0.07
	PHATE, 2 dims	8	0.231	0.12	0.04	0.07
	t-SNE, 2 dims	8	0.585	0.13	0.07	0.09
	SR-tSNE, 2 dims	8	0.590	0.13	0.07	0.09
LoCoMo	UMAP, 16 dims	64	0.362	0.19	0.30	0.08
	PCA to 50, UMAP to 16	64	0.329	0.19	0.25	0.07
	PHATE, 16 dims	64	0.292	0.22	0.24	0.08
	UMAP, 2 dims	8	0.255	0.16	0.21	0.04
	PHATE, 2 dims	8	0.141	0.19	0.14	0.05
	t-SNE, 2 dims	8	0.388	0.20	0.15	0.05
	SR-tSNE, 2 dims	8	0.392	0.20	0.15	0.05

TABLE 9
At equal bytes, compressions of the financial memory differ most in G and theme agreement. FinQA, BGE-base, 27 compressions: bytes per item, fidelity, and agreement with the uncompressed memory. Retrieval: overlap of the top-10 lists for 863 questions. Evidence: evidence recall@10 (0.50 uncompressed). Themes: ARI of consolidation into 11 themes. Peers: overlap of five-company peer groups. Barrier: Jaccard overlap of the memories removed to wall off a company. Map: rank correlation of distances among company centroids.

Compression	Bytes	L	G	Retrieval	Evidence	Themes	Peers	Barrier	Map
float16	1536	1.00	1.00	1.00	0.50	1.00	1.00	1.00	1.00
int8	768	0.98	0.97	0.97	0.50	0.99	0.99	0.98	1.00
int4	384	0.96	0.95	0.94	0.50	0.99	0.99	0.97	1.00
binary, 1 bit	96	0.74	0.65	0.57	0.44	0.79	0.80	0.87	0.97
binary, 2 bit	192	0.83	0.65	0.70	0.48	0.90	0.83	0.89	0.97
binary, 4 bit	384	0.86	0.72	0.76	0.49	0.94	0.85	0.90	0.97
binary + rescoring	96	0.97	0.65	0.87	0.49	0.79	0.80	0.87	0.97
truncation 384	1536	0.86	0.76	0.74	0.49	0.77	0.75	0.89	0.96
truncation 192	768	0.76	0.62	0.58	0.46	0.62	0.59	0.81	0.90
truncation 96	384	0.64	0.47	0.40	0.33	0.44	0.46	0.66	0.80
truncation 48	192	0.47	0.33	0.22	0.21	0.22	0.31	0.49	0.64
PCA 384	1536	0.97	0.97	0.96	0.49	1.00	1.00	0.98	1.00
PCA 192	768	0.89	0.86	0.81	0.44	0.99	0.92	0.92	0.99
PCA 96	384	0.77	0.74	0.60	0.35	0.96	0.78	0.84	0.95
PCA 48	192	0.63	0.65	0.39	0.24	0.92	0.61	0.60	0.88
random projection 384	1536	0.80	0.66	0.64	0.47	0.67	0.67	0.82	0.91
random projection 192	768	0.73	0.57	0.51	0.39	0.59	0.59	0.72	0.87
random projection 96	384	0.62	0.45	0.37	0.37	0.34	0.40	0.57	0.73
random projection 48	192	0.45	0.35	0.20	0.18	0.21	0.33	0.37	0.67
PCA 192 + int8	192	0.88	0.84	0.80	0.43	0.99	0.93	0.92	0.99
PCA 96 + int8	96	0.77	0.74	0.60	0.35	0.96	0.78	0.84	0.95
PCA 768 + rotation + binary	96	0.77	0.57	0.62	0.45	0.85	0.82	0.88	0.96
PCA 384 + rotation + binary	48	0.72	0.51	0.52	0.41	0.79	0.82	0.88	0.96
PQ 96	96	0.82	0.73	0.68	0.44	0.88	0.87	0.90	0.98
PQ 48	48	0.71	0.62	0.49	0.33	0.79	0.78	0.85	0.95
PQ 24	24	0.60	0.51	0.32	0.20	0.52	0.65	0.78	0.89
UMAP 16	64	0.52	0.10	0.24	0.19	0.08	0.20	0.69	0.59